

Homework

Preliminary Mathematics Standard

Data Analysis

1 Questions

1.1 Descriptive Statistics

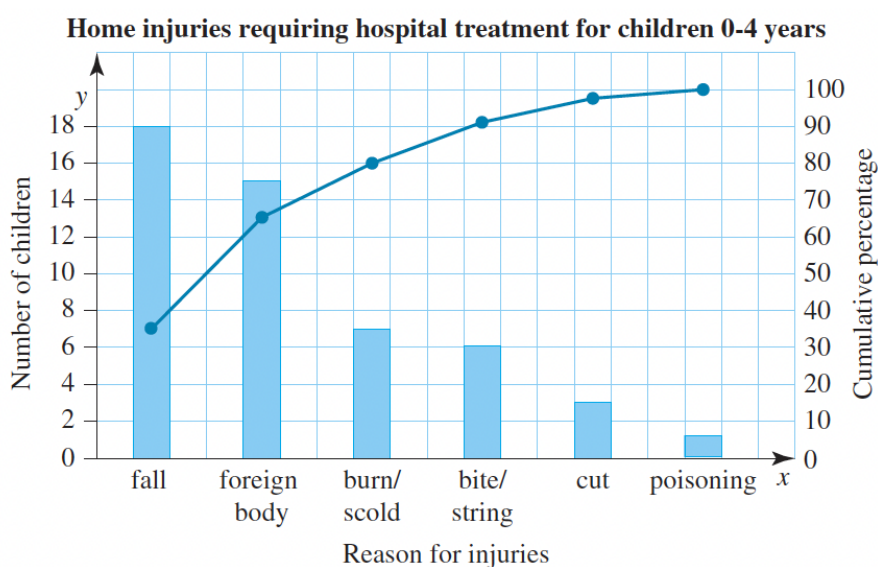
1. A survey was conducted asking members of the public to nominate their preferred sport. The results recorded were:

football, cricket, cricket, tennis, basketball, netball, tennis, netball, swimming, netball, tennis, football, cricket, basketball, lawn bowls, football, swimming, netball, tennis, netball, cricket, tennis, football, basketball, swimming, lawn bowls, swimming, swimming, netball, netball, tennis, golf, football, football, basketball, swimming, golf, football, netball, swimming, basketball, basketball, golf, tennis, cricket, cricket, football, basketball, netball, golf.

- (a) Is the data collected in this survey an example of numerical or categorical data?
 - (b) What problems arise when the results of surveys are presented as they are above instead of in table form?
 - (c) Present the data in a frequency table and use the table to write a description summarising the survey results.
2. (a) List two examples of your own that would be considered discrete data.
(b) List two examples of your own that would be considered continuous data.
 3. Students from a Year 11 class were asked about their favourite subjects and the following data was recorded.

Maths	English	PE	Science	PE
Art	Maths	Science	English	Science
Cooking	PE	English	Cooking	Maths
PE	Art	Science	Maths	Art
Science	Cooking	Art	PE	PE

- (a) Put the data in a frequency table and record the number of students who were surveyed.
 - (b) What percentage of students preferred Maths?
 - (c) What was the most popular subject? What percentage of students preferred this?
 - (d) What type of data is this?
4. The following Pareto chart shows the causes of home injuries for which children aged 0 to 4 years required hospital treatment in April 2017.

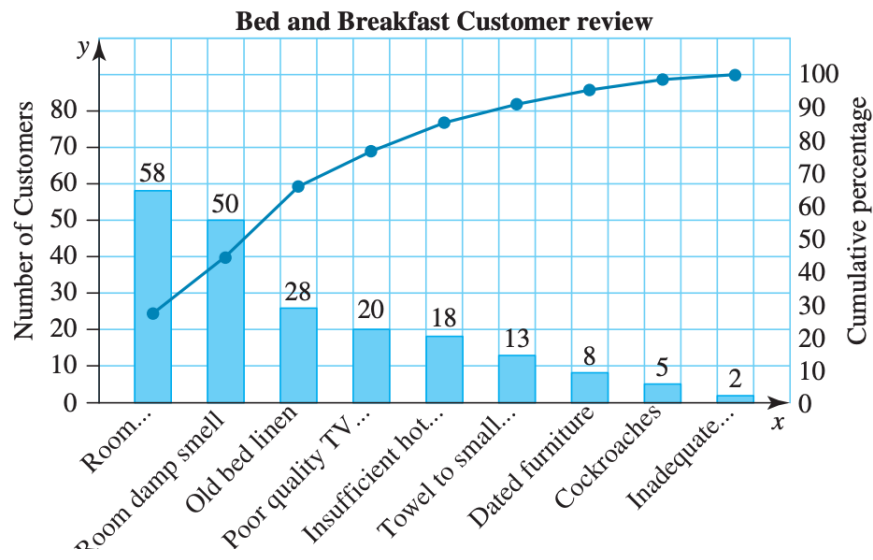


- (a) How many children required hospital treatment in April 2017 for home injuries?
- (b) Which reason for home injury for children aged 0 to 4 years is the biggest concern?
- (c) What is the cumulative percentage frequency value for burn/scald injuries?

1.2 Histograms & Polygons

1. An online bed and breakfast company gives customers the opportunity to review online the bed and breakfast places that they stayed in. The table and Pareto chart below show the complaints of customers staying at a particular bed and breakfast place located in an alpine resort area.
 - (a) How many customers reviewed the bed and breakfast place?

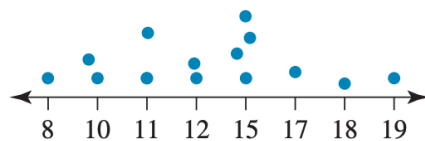
Customer reasons	Frequency
Too cold	58
Damp smell	50
Old bed linen	26
Poor TV reception	20
Insufficient hot water	18
Towels too small/thin	13
Dated furniture	8
Cockroaches	5
Inadequate lighting	2



- (b) Which reason for customer complaints is the biggest concern for the bed and breakfast?
 - (c) Is this sampling method used by the online bed and breakfast company biased? Explain.
2. The following diagram is the result of a student's attempt to draw a dot plot to display the values

8, 10, 10, 11, 11, 12, 12, 15, 15, 15, 15, 17, 18, 19.

- (a) List the mistakes that the student made in drawing the dot plot.
- (b) Draw the correct dot plot.



3. The maximum daily temperatures ($^{\circ}\text{C}$) in Sydney for the month of February were recorded as follows:

28, 24, 29, 37, 35, 30, 31, 27, 34, 29, 25, 36, 36, 37, 36, 30, 42, 41, 42, 41, 38, 37, 33, 27, 26, 36, 27, 26.

- Draw a dot plot to display the data.
 - Comment on the distribution of the data.
4. Train timetables at railway stations are often presented in a format similar to stem-and-leaf plots. A section of a timetable is displayed below.

Table 1: Departure times from Flinders Street station

<i>am</i>	
6	03 23 43
7	03 18 33 48
8	03 10 18 25 33 43 50
9	03 18 33 48

- What does this timetable use as the stem and the leaf?
 - If you arrive at the station at 8:20, what train will you take?
 - What benefit does a timetable in this form have compared with a timetable that lists the full time for each departure?
5. Data was collected on the number of times people go to the cinema in a month.

4, 5, 7, 9, 1, 2, 5, 2, 4, 8, 3, 6,
2, 3, 8, 1, 1, 4, 5, 3, 3, 6, 1, 2,
7, 1, 3, 2, 2, 4, 10, 0, 1, 3, 4, 6

- Put the data in a frequency table.
- Draw a histogram to represent the data.
- How many people go to the cinema less than three times a month?
- How many people go to the cinema at least three times a month?

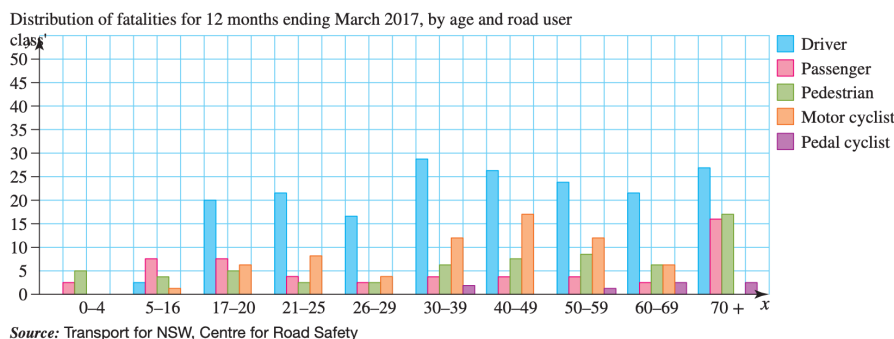
6. The daily numbers of hits a fashion blogger gets on her new website over 3 weeks are:

126 356 408 404 420 425 176
 167 398 433 446 419 431 189
 120 431 390 495 454 215 11z

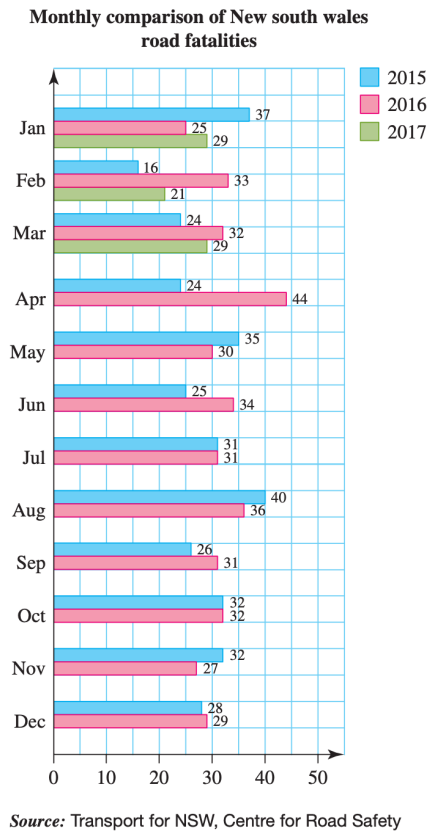
At the same time, the daily number of hits a healthy lifestyle blogger gets on his new website over 3 weeks are:

240 156 462 510 420 474 520
 225 402 426 563 621 339 195
 320 621 340 495 700 415 371

- (a) Compare the two data sets using an appropriate graphical display.
 (b) Comment on the two data sets.
7. A side-by-side bar chart shows the distributions of New South Wales road fatalities from March 2016 to March 2017.



- (a) Which age group had the most passenger and pedestrian fatalities? Give a reason why this might be.
- (b) The New South Wales government wants to introduce a road campaign. Which age groups should the government focus on? Explain your answer.
8. A horizontal side-by-side bar chart shows a monthly comparison of the New South Wales road fatalities.
- (a) Using the data from 2015 and 2016, which year had the most fatalities on New South Wales roads?
- (b) Which year has had the most fatalities from January to March?
- (c) Why do you think there were 44 fatalities in the month of April 2016?



1.3 Measures of Centre and Spread

- The ages of 9 employees at a small business are:

24, 31, 28, 45, 31, 52, 28, 31, 38

- Calculate the mean age.
 - Find the median age.
 - State the mode.
 - Calculate the range.
- The following frequency table shows the number of hours per week students spend on social media.

Hours per week	Frequency
5	4
10	9
15	12
20	7
25	3

- How many students were surveyed?
 - Calculate the mean number of hours spent on social media per week.
 - Find the modal number of hours.
 - Find the median number of hours.
3. Two basketball players record their points scored over 8 games:

Player A: 12, 15, 14, 13, 16, 12, 15, 11

Player B: 8, 22, 10, 25, 6, 19, 14, 24

- Calculate the mean score for each player.
 - Calculate the range for each player.
 - Calculate the population standard deviation for each player, correct to 2 decimal places.
 - Which player would you select for a team that needs a *consistent* scorer? Justify your answer.
4. A dataset has a mean of 18, a median of 14 and a mode of 11.
- Which measure of centre would best represent the typical value in this dataset? Justify your answer.
 - Suggest a possible reason why the mean is so much higher than the median and mode.
 - Give a real-world context where this pattern of mean > median > mode commonly occurs.
5. The daily maximum temperatures (in °C) recorded over two weeks in Sydney were:

28, 31, 33, 29, 35, 30, 27, 26, 32, 34, 31, 29, 33, 30

- Calculate the mean temperature, correct to 1 decimal place.
- Find the median temperature.
- Calculate the range.
- Calculate the population standard deviation, correct to 2 decimal places.

- (e) Describe the distribution shape of this dataset.
6. A factory produces bolts. A quality control officer measures the diameter (mm) of a sample of 10 bolts:

10.1, 10.3, 9.8, 10.2, 10.0, 10.4, 9.9, 10.1, 10.2, 10.0

- (a) Calculate the mean diameter.
- (b) Calculate the population standard deviation, correct to 3 decimal places.
- (c) A second machine produces bolts with a mean diameter of 10.1 mm and a standard deviation of 0.25 mm. Which machine produces more consistent bolts? Justify your answer.
- (d) Why is consistency important in manufacturing?
7. The stem-and-leaf plot below shows the marks (out of 50) scored by students on a test.

Stem	Leaf
2	3 5 8
3	1 1 4 7 9
4	0 2 2 5 5 5 8
5	0 0

- (a) How many students sat the test?
- (b) Write out the full dataset in order.
- (c) Find the median mark.
- (d) State the mode.
- (e) Calculate the range.
- (f) Describe the shape of this distribution.
8. The annual salaries of employees at a company are:
- \$52 000, \$55 000, \$58 000, \$61 000, \$63 000, \$410 000
- (a) Calculate the mean salary.
- (b) Find the median salary.
- (c) A journalist reports that the “average” salary at the company is over \$100 000. Which measure of centre are they using, and is this a fair representation? Explain.
- (d) Which measure of centre best represents the typical salary? Why?
9. The number of goals scored per match by a soccer team over a season is shown in the table below.

Goals scored	Frequency	Cumulative frequency
0	3	
1	7	
2	10	
3	6	
4	4	

- Complete the cumulative frequency column.
 - How many matches did the team play?
 - Calculate the mean number of goals per match.
 - Use the cumulative frequency column to find the median.
 - State the mode.
 - Calculate the population standard deviation, correct to 2 decimal places.
10. A dataset of 7 values has the following properties:

$$\text{Mean} = 12, \quad \text{Mode} = 10, \quad \text{Range} = 14$$

The dataset contains the values 10, 10, 8, 15, 22 and two unknown values a and b , where $a < b$.

- Use the mean to write an equation involving a and b .
- Use the range to write a second equation or inequality involving a and b .
- Hence find the values of a and b .
- Calculate the population standard deviation of the complete dataset, correct to 2 decimal places.

1.4 Quartiles, Box Plots and Outliers

- Find the five-number summary and IQR for the following dataset:

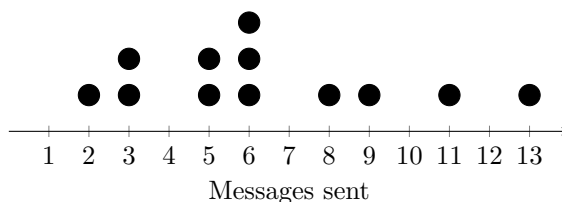
4, 7, 9, 12, 15, 18, 21, 24, 27

- The ages of 10 people at a community event were recorded as:

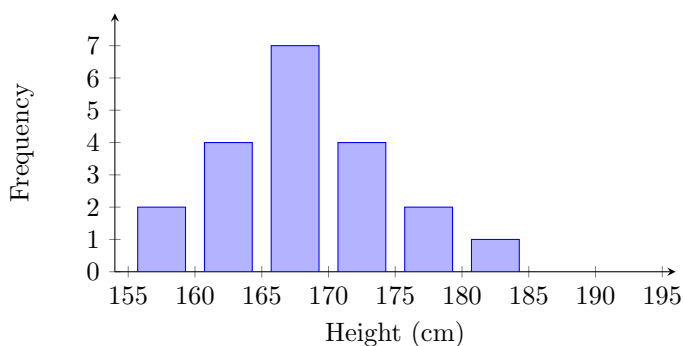
14, 19, 22, 25, 25, 31, 34, 38, 45, 52

- Find the five-number summary.
- Calculate the IQR.
- Calculate the range.
- Which measure of spread — range or IQR — better represents this dataset? Justify your answer.

3. The dot plot below shows the number of text messages sent by 12 students in one day.



- Write out the dataset in order.
 - Determine the five-number summary.
 - Draw a box plot to represent this data.
4. The following frequency histogram shows the heights (cm) of 20 students.



- How many students are in the dataset?
 - Determine the five-number summary.
 - Construct a box plot for this data.
 - Describe the shape of the distribution.
5. The following dataset contains an error made during data collection:

11, 13, 14, 15, 15, 16, 17, 18, 82

- Calculate Q_1 , Q_3 and the IQR.
- Use the outlier formula to determine whether 82 is a formal outlier.
- Calculate the mean with and without the outlier.
- Calculate the median with and without the outlier.
- What do your results suggest about the effect of outliers on the mean versus the median?

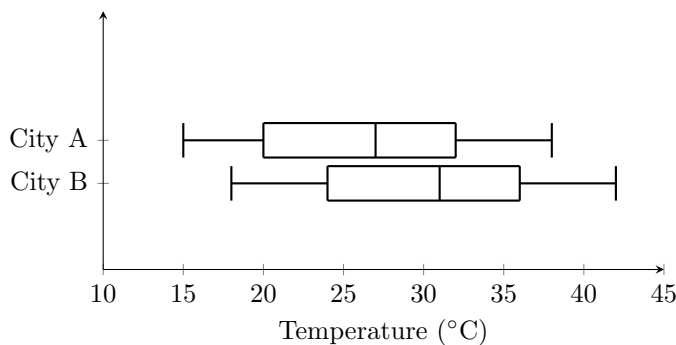
6. The results of a class spelling test (out of 20) are shown below:

8, 10, 11, 12, 13, 13, 14, 14, 15, 15, 15, 16, 17, 18, 19

- Find the five-number summary.
 - Draw a box plot to represent this data on a clearly labelled number line.
 - Identify any clusters or gaps in the data and explain what they might suggest.
 - Are there any outliers? Show your working using the formal outlier rule.
7. Two rugby teams record their points scored across a season. The five-number summaries are given below.

	Min	Q_1	Median	Q_3	Max
Team A	6	14	22	28	38
Team B	10	18	24	30	44

- Draw a parallel box plot for both teams on the same number line.
 - Calculate the IQR for each team.
 - Compare the centres of the two distributions.
 - Compare the spreads of the two distributions.
 - Which team is more consistent? Justify your answer.
 - Describe the shape of each distribution from the box plot.
8. The following parallel box plots show the daily maximum temperatures ($^{\circ}\text{C}$) recorded in two cities over one month.



- State the median temperature for each city.
- Calculate the IQR for each city.

- (c) Which city had a higher typical temperature? Justify using statistics.
- (d) Which city had more variable temperatures? Justify your answer.
- (e) Compare the shapes of the two distributions.

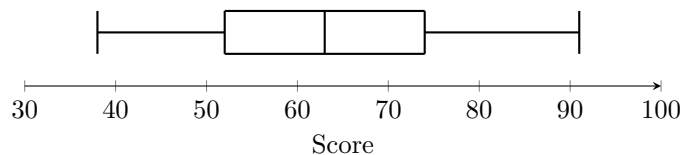
9. A dataset of house prices (\$'000s) in a suburb is:

420, 445, 460, 475, 480, 490, 505, 510, 520, 1250

- (a) Find the five-number summary.
 - (b) Use the formal outlier rule to determine whether \$1 250 000 is an outlier.
 - (c) Calculate the mean and median with the outlier included.
 - (d) Calculate the mean and median with the outlier excluded.
 - (e) A real estate agent advertises the “average” house price using the mean including the outlier. Is this misleading? Justify your answer.
 - (f) Which measure of centre is more appropriate here? Why?
10. The times (in minutes) taken by 14 runners to complete a 10 km race were:

38, 41, 44, 44, 46, 47, 49, 51, 52, 54, 55, 58, 61, 90

- (a) Find the five-number summary.
 - (b) Formally test whether 90 minutes is an outlier.
 - (c) Draw a box plot, marking any outliers as separate points.
 - (d) Describe the shape of the distribution.
 - (e) Identify any clusters or gaps. Suggest a real-world reason for the outlier.
11. The box plot below represents the scores of 30 students on a maths exam.



- (a) State the five-number summary.
- (b) Calculate the IQR and range.
- (c) What percentage of students scored between 52 and 74?
- (d) What percentage of students scored above 63?
- (e) The pass mark is 50. Estimate the percentage of students who passed.

- (f) Describe the shape of the distribution and what it suggests about student performance.
12. The number of daily visitors to two websites over 11 days were recorded:
- Site A:** 120, 135, 140, 142, 145, 148, 150, 153, 155, 160, 162
- Site B:** 80, 95, 110, 130, 145, 150, 155, 170, 185, 200, 310
- Find the five-number summary for each site.
 - Calculate the IQR and range for each site.
 - Formally test whether 310 is an outlier for Site B.
 - Draw a parallel box plot for both sites, marking any outliers.
 - Compare the centre, spread and shape of the two distributions.
 - Which site has more predictable daily traffic? Justify your answer.

13. A dataset has the following five-number summary:

$$\text{Min} = 12, \quad Q_1 = 20, \quad Q_2 = 28, \quad Q_3 = 35, \quad \text{Max} = 60$$

- Calculate the IQR.
- Use the formal outlier rule to find the upper and lower fences.
- Is the maximum value of 60 a formal outlier?
- Sketch the box plot, marking any outliers appropriately.
- Describe the shape of the distribution from the box plot.
- Explain how removing the outlier (if any) would affect the mean and standard deviation.

1.5 Statistical Investigation Process and Sampling

- A student wants to investigate the sleeping habits of teenagers in Australia. She decides to post a survey on her personal social media page and record the responses.
 - What is the population for this investigation?
 - What sampling method is she using?
 - Give two reasons why her sample may not be representative of the population.
 - Rewrite the following survey question to make it clear and unambiguous:
“Do you sleep enough and wake up feeling okay?”
 - The survey asks respondents to select their age group from: “young, middle-aged, old.” Identify two problems with these response options and suggest an improvement.

2. A school of 900 students is divided into three year groups: 300 in Year 10, 360 in Year 11 and 240 in Year 12. A researcher wants to take a stratified sample of 90 students.
 - (a) Explain why stratified sampling is more appropriate than simple random sampling for this investigation.
 - (b) Calculate how many students should be selected from each year group.
 - (c) Describe how the researcher would then select the students from each year group.
3. For each of the following scenarios, identify the sampling method being used and state one advantage and one disadvantage of that method.
 - (a) A supermarket surveys every 15th customer who enters the store on a Saturday morning.
 - (b) A radio station asks listeners to call in and vote on whether a new law should be introduced.
 - (c) A researcher assigns every employee in a company a number and uses a random number table to select 40 of them.
 - (d) A hospital divides its patients into age groups and randomly selects a proportional number from each group for a health survey.
4. Read the following survey questions and identify the problem with each. Then rewrite each question to correct the issue.
 - (a) “Don’t you think the government should spend more money on public transport?”
 - (b) “How many hours do you spend exercising and cooking healthy meals each week?”
 - (c) “Do you sometimes occasionally skip breakfast sometimes?”
 - (d) “Rate your income: low / okay / high.”
5. A researcher is investigating whether drinking coffee improves study performance in university students. She observes students in a library and notes whether they have a coffee cup and how long they study for.
 - (a) What type of data collection method is being used?
 - (b) Identify one potential confounding variable in this study.
 - (c) Explain why the researcher cannot conclude that coffee *causes* students to study longer.
 - (d) Suggest a more controlled method the researcher could use to better investigate this question.
6. A factory produces 2 400 phone cases per day. A quality inspector wants to check a sample of 80 cases for defects using systematic sampling.

- (a) Calculate the sampling interval.
 - (b) Describe how the inspector would carry out this sampling method.
 - (c) Suggest one situation in which systematic sampling could produce a biased sample in this context.
 - (d) If it was found that defective cases tend to occur in groups of 30 during a shift change, which sampling method would be more appropriate? Justify your answer.
7. Two newspaper headlines report on the same dataset about average weekly wages:
- Headline A:** *“Average wage rises to \$1 450 — workers better off than ever!”*
- Headline B:** *“Half of all workers earn less than \$1 100 a week”*
- (a) Which measure of centre is Headline A most likely using? Which is Headline B using?
 - (b) Suggest why the two headlines give such different impressions of the same data.
 - (c) What does the difference between these two values suggest about the shape of the income distribution?
 - (d) Which headline gives a more accurate picture of the typical worker’s wage? Justify your answer.
8. A local council wants to find out whether residents support the construction of a new sports centre. They are considering three data collection approaches:
- **Option A:** Mail a survey to every 10th household on the electoral roll
 - **Option B:** Set up a booth at the shopping centre and invite passers-by to respond
 - **Option C:** Post a survey link on the council’s Facebook page
- (a) Identify the sampling method used in each option.
 - (b) For each option, describe one way in which the sample may not represent the full population of residents.
 - (c) Which option would produce the most representative sample? Justify your answer.
 - (d) Suggest one ethical consideration the council should address in any of these surveys.
9. A study reports: *“Suburbs with more ice cream shops have higher rates of drowning deaths. Therefore, ice cream causes drowning.”*

- (a) Identify the flaw in the study's conclusion.
 - (b) Suggest a variable that could explain the relationship between ice cream sales and drowning rates.
 - (c) What is the term for a variable that influences both the explanatory and response variables in a study?
 - (d) Give one other real-world example of two variables that are correlated but where one does not cause the other.
10. A researcher is designing a survey to investigate the dietary habits of Australian adults aged 18–65. She plans to survey 200 people.
- (a) Identify one issue of privacy the researcher should consider and explain how she could address it.
 - (b) The researcher plans to conduct the survey only at a gym on week-day mornings. Identify two groups of the population who would be underrepresented and explain the impact this would have on the results.
 - (c) Write two survey questions — one about fruit consumption and one about fast food consumption — that follow good questionnaire design principles. Each question must include response options.
 - (d) The researcher finds that 85% of respondents eat fruit daily and concludes that “most Australian adults have a healthy diet.” Identify two problems with this conclusion.

2 Answers

2.1 Descriptive Statistics

1. (a) Categorical
- (b) Very difficult to accurately analyse the results
- (c)

Sport	Frequency
Netball	9
Football	8
Tennis	7
Basketball	7
Swimming	7
Cricket	6
Golf	4
Lawn bowls	2

The popularity of most sports was evenly spread, with the exception of the low-scoring lawn bowls and golf.

2. Answers will vary. Examples are given.
 - (a) Number of people in a cinema; number of security boxes at a bank
 - (b) Weights of students in your class; distance each student lives from school
3. (a)

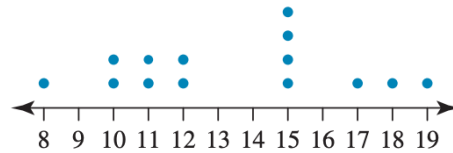
Subject	Tally	Frequency (f)
Maths		4
Art		4
Cooking		3
PE	 	6
Science	 	5
English		3
Total		25

- (b) 16%

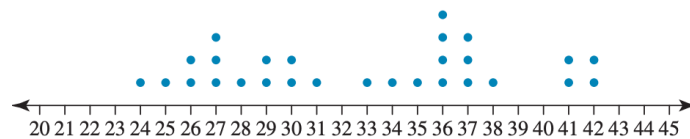
- (c) PE; 24%
 - (d) Nominal (categorical)
4. (a) 50
- (b) Fall
 - (c) 80%

2.2 Histograms & Polygons

1. (a) 200
- (b) Room temperature too cold
 - (c) Answers will vary. The sampling method used by the bed and breakfast is biased as the online reviews are voluntary. Guests can submit multiple reviews, and dissatisfied guests are more likely to write a review and give their opinion.
2. (a) Dots are not aligned vertically. The horizontal scale is not equally sapaced.



- (b)
3. 12



- (a)
 - (b) There is a wide spread of temperatures.
4. (a) The stem represents the hour and the leaves represent the minutes.
- (b) 8:25 am
 - (c) This type of timetable enables the viewer to see times at a glance, and it saves space.
5. (a)
- (b)

- (c) 13
 - (d) 23
6. (a)
- (b) Answers will vary. Example answer: The fashion blogger had 400–450 hits on her website on 9 days in the 3-week period. The healthy life style blogger got a wider spread of hits. He received over 400 hits for 11 days in the 3-week period. His website seems to be gaining in popularity.
7. (a) 70 + age group. Answers about the reason will vary. Elderly people don't drive as much and tend to walk or be passengers.
- (b) Answers will vary. The road campaign could focus on:
- elderly people as pedestrians — being alert when crossing a road.
 - motorcyclists in the 30–49 age group, not on probationary license — risk taking
 - driving for all age groups — awareness and safety for all drivers.
8. (a) 2016
- (b) 2016
- (c) Answers will vary but could include that more people are on the roads during the Easter break and school holidays.

2.3 Measures of Centre and Spread

1. (a) Mean = 34.2 years
 - (b) Median = 31
 - (c) Mode = 31
 - (d) Range = 28
2. (a) $n = 35$
 - (b) Mean = 14.43 hours
 - (c) Mode = 15 hours
 - (d) Median = 15 hours
3. (a) Player A: mean = 13.5; Player B: mean = 16
 - (b) Player A: range = 5; Player B: range = 19
 - (c) Player A: $\sigma \approx 1.58$; Player B: $\sigma \approx 6.87$
 - (d) Player A — lower standard deviation indicates more consistent scoring
4. (a) Median (= 14) — less affected by the outlier(s) pulling the mean up
5. (a) Mean $\approx 30.6^\circ\text{C}$

- (b) Median = 30.5°C
 - (c) Range = 9°C
 - (d) $\sigma \approx 2.37^{\circ}\text{C}$
 - (e) Approximately unimodal and roughly symmetric
6. (a) Mean = 10.1 mm
- (b) $\sigma \approx 0.167$ mm
 - (c) First machine ($\sigma \approx 0.167$ mm $<$ 0.25 mm)
7. (a) $n = 17$
- (b) 23, 25, 28, 31, 31, 34, 37, 39, 40, 42, 42, 45, 45, 45, 48, 50, 50
 - (c) Median = 42
 - (d) Mode = 45
 - (e) Range = 27
 - (f) Approximately unimodal, slightly negatively skewed
8. (a) Mean = \$116 500
- (b) Median = \$59 500
 - (c–d) Median (= \$59 500) — the \$410 000 salary is an outlier that heavily inflates the mean
9. (a) 3, 10, 20, 26, 30
- (b) $n = 30$ matches
 - (c) Mean = 2.03 goals
 - (d) Median = 2 goals
 - (e) Mode = 2 goals
 - (f) $\sigma \approx 1.13$
10. (a) $a + b = 19$
- (b) $b = 22$ or $b \leq 22$
 - (c) $a = 6$, $b = 13$
 - (d) $\sigma \approx 4.27$

2.4 Quartiles, Box Plots and Outliers

1. (a) Min = 4, $Q_1 = 7$, Median = 15, $Q_3 = 24$, Max = 27
- (b) IQR = $24 - 7 = 17$
2. (a) Min = 14, $Q_1 = 22$, Median = 28, $Q_3 = 38$, Max = 52
- (b) IQR = $38 - 22 = 16$
- (c) Range = $52 - 14 = 38$

- (d) The IQR is more appropriate — the range is heavily influenced by the extreme values of 14 and 52, whereas the IQR describes the spread of the middle 50% of the data more reliably.
3. (a) 2, 3, 3, 5, 5, 6, 6, 6, 8, 9, 11, 13
 (b) $\text{Min} = 2$, $Q_1 = 4$, $\text{Median} = 6$, $Q_3 = 8.5$, $\text{Max} = 13$
 (c) Box plot with whiskers from 2 to 4, box from 4 to 8.5, median line at 6, whisker to 13.
4. (a) $n = 20$
 (b) $\text{Min} = 155$, $Q_1 = 165$, $\text{Median} = 167.5$, $Q_3 = 172.5$, $\text{Max} = 185$
 (c) Box plot with whiskers from 155 to 165, box from 165 to 172.5, median line at 167.5, whisker to 185.
 (d) Approximately unimodal and slightly positively skewed — most values are clustered in the 165–170 cm range with a longer tail to the right.
5. (a) $Q_1 = 13$, $Q_3 = 17.5$, $\text{IQR} = 4.5$
 (b) Lower fence $= 13 - 1.5 \times 4.5 = 6.25$; Upper fence $= 17.5 + 1.5 \times 4.5 = 24.25$. Since $82 > 24.25$, the value 82 is a formal outlier.
 (c) Mean with outlier ≈ 22.3 ; mean without outlier ≈ 14.3
 (d) Median with outlier $= 15$; median without outlier $= 14.5$
 (e) The mean increased by 8 when the outlier was included, while the median barely changed. This confirms that the median is resistant to outliers whereas the mean is not.
6. (a) $\text{Min} = 8$, $Q_1 = 12$, $\text{Median} = 15$, $Q_3 = 17$, $\text{Max} = 19$
 (b) Box plot with whiskers from 8 to 12, box from 12 to 17, median line at 15, whisker to 19.
 (c) The scores cluster between 13 and 16, suggesting most students performed at a similar level. There is a gap between 8 and 10, possibly indicating a small group of students who were underprepared.
 (d) $\text{IQR} = 17 - 12 = 5$. Lower fence $= 12 - 7.5 = 4.5$; Upper fence $= 17 + 7.5 = 24.5$. No values fall outside these fences, so there are **no outliers**.
7. (a) Parallel box plot drawn with both teams on the same scale from 0 to 50.
 (b) $\text{IQR}_A = 28 - 14 = 14$; $\text{IQR}_B = 30 - 18 = 12$
 (c) Team B has a higher median (24 vs 22), indicating a slightly higher typical score.
 (d) Team B has a slightly smaller IQR (12 vs 14), suggesting marginally more consistency in the middle 50% of scores. However, Team B has a larger overall range (34 vs 32).

- (e) Team B is more consistent based on the smaller IQR.
 - (f) Team A: median is slightly left of centre in the box, suggesting mild positive skew. Team B: median is approximately centred, suggesting a roughly symmetric distribution.
8. (a) City A: Median = 27°C ; City B: Median = 31°C
- (b) City A: $\text{IQR} = 32 - 20 = 12$; City B: $\text{IQR} = 36 - 24 = 12$
- (c) City B had a higher typical temperature — its median of 31°C is 4°C higher than City A's median of 27°C .
- (d) Both cities have the same IQR of 12, but City B has a larger range ($42 - 18 = 24$ vs $38 - 15 = 23$), so City B is very slightly more variable overall.
- (e) Both distributions appear approximately symmetric — the median sits near the centre of each box and the whiskers are of similar length.
9. (a) Min = 420, $Q_1 = 460$, Median = 487.5, $Q_3 = 512.5$, Max = 1250
- (b) $\text{IQR} = 512.5 - 460 = 52.5$. Upper fence = $512.5 + 1.5 \times 52.5 = 591.25$. Since $1250 > 591.25$, \$1 250 000 is a formal outlier.
- (c) Mean (with) $\approx \$555\,500$; Median (with) = \$487 500
- (d) Mean (without) $\approx \$478\,333$; Median (without) = \$482 500
- (e) Yes — the mean of \$555 500 is misleading as it is \$77 167 above what most houses sell for. The single \$1 250 000 sale pulls the mean up significantly.
- (f) The median is more appropriate as it is resistant to the outlier and better reflects the typical house price in the suburb.
10. (a) Min = 38, $Q_1 = 44$, Median = 50, $Q_3 = 55$, Max = 90
- (b) $\text{IQR} = 55 - 44 = 11$. Upper fence = $55 + 1.5 \times 11 = 71.5$. Since $90 > 71.5$, 90 minutes is a formal outlier.
- (c) Box plot drawn from 38 to 68 (largest non-outlier), with 90 plotted as a separate point (*).
- (d) Positively skewed — most runners finished between 38 and 60 minutes, with one unusually slow time pulling the distribution to the right.
- (e) Cluster between 44 and 58 minutes (the bulk of the field). A gap exists between 61 and 90 minutes. The outlier at 90 minutes may represent a runner who was injured or stopped during the race.
11. (a) Min = 38, $Q_1 = 52$, Median = 63, $Q_3 = 74$, Max = 91
- (b) $\text{IQR} = 74 - 52 = 22$; Range = $91 - 38 = 53$
- (c) The middle 50% of students (between Q_1 and Q_3) scored between 52 and 74, so 50% of students scored in that range.
- (d) 50% of students scored above the median of 63.

- (e) $Q_1 = 52 > 50$, so approximately 75% of students passed.
 - (f) The distribution is approximately symmetric — the median sits near the centre of the box and the whiskers are of similar length. This suggests most students achieved moderate scores with few extreme results.
12. (a) Site A: Min = 120, $Q_1 = 140$, Median = 148, $Q_3 = 155$, Max = 162
 Site B: Min = 80, $Q_1 = 110$, Median = 150, $Q_3 = 185$, Max = 310
- (b) Site A: IQR = 15, Range = 42; Site B: IQR = 75, Range = 230
 - (c) $IQR_B = 75$. Upper fence = $185 + 1.5 \times 75 = 297.5$. Since $310 > 297.5$, 310 **is** a formal outlier for Site B.
 - (d) Parallel box plot drawn with Site B's outlier marked separately.
 - (e) Both sites have similar medians (≈ 148 vs 150), so their typical daily traffic is comparable. However, Site B has a much larger IQR (75 vs 15) and range (230 vs 42), indicating far more variable traffic. Site A is approximately symmetric; Site B is positively skewed due to the outlier.
 - (f) Site A — its much smaller IQR and range indicate that daily visitor numbers are far more consistent and predictable.
13. (a) $IQR = 35 - 20 = 15$
- (b) Lower fence = $20 - 1.5 \times 15 = -2.5$; Upper fence = $35 + 1.5 \times 15 = 57.5$
 - (c) Since $60 > 57.5$, the maximum value of 60 **is** a formal outlier.
 - (d) Box plot drawn with whiskers from 12 to 20 and 35 to the largest non-outlier, with 60 marked as a separate point.
 - (e) The distribution is positively skewed — the median sits closer to Q_1 than Q_3 , and there is an outlier on the upper end.
 - (f) Removing the outlier would decrease the mean (as it was inflated by the high value) and decrease the standard deviation (as values would be less spread from the mean). The median would be largely unaffected.

2.5 Statistical Investigation Process and Sampling

1. (a) All teenagers in Australia.
- (b) Self-selected (voluntary response) sampling.
- (c) The sample is limited to people who follow her on social media, excluding teenagers without social media access. People with strong opinions are more likely to respond, causing voluntary response bias.
- (d) “How many hours of sleep do you get on a typical school night?”
 - Less than 6 hours 6–7 hours 8–9 hours More than 9 hours

- (e) The terms “young”, “middle-aged” and “old” are subjective and mean different things to different people. There are no defined age boundaries. An improvement would be to use specific numeric ranges, e.g. 13–17, 18–25, 26–35, 36–50, 51–65, 65+.
2. (a) Stratified sampling ensures each year group is represented in the sample in proportion to its size in the school. Simple random sampling might, by chance, over- or under-represent one or more year groups.
- (b)
- $$\text{Year 10: } \frac{300}{900} \times 90 = 30 \text{ students}$$
- $$\text{Year 11: } \frac{360}{900} \times 90 = 36 \text{ students}$$
- $$\text{Year 12: } \frac{240}{900} \times 90 = 24 \text{ students}$$
- (c) Within each year group, assign every student a number and use a random number generator (or draw names) to select the required number of students.
3. (a) Systematic sampling. Advantage: simple and quick to apply. Disadvantage: could introduce bias if customer arrival patterns repeat (e.g. every 15th customer is always a staff member).
- (b) Self-selected sampling. Advantage: easy and inexpensive to conduct. Disadvantage: only people with strong opinions tend to respond, making results unrepresentative.
- (c) Random sampling. Advantage: every employee has an equal chance of selection, minimising bias. Disadvantage: requires a complete list of all employees.
- (d) Stratified sampling. Advantage: ensures all age groups are proportionally represented. Disadvantage: more complex and time-consuming to organise than simple random sampling.
4. (a) Leading question — the phrasing pressures the respondent to agree.
Rewrite: “What is your view on government spending on public transport?”
- Should increase Should stay the same Should decrease Unsure
- (b) Double-barrelled — asks about two separate behaviours in one question.
Rewrite: Split into two questions: “How many hours per week do you spend exercising?” and “How many hours per week do you spend cooking meals at home?”
- (c) Ambiguous and poorly worded — “sometimes occasionally” is redundant and unclear.
Rewrite: “How many days per week do you skip breakfast?”

- Never 1–2 days 3–4 days 5–7 days
- (d) The terms “low”, “okay” and “high” are subjective with no defined boundaries.
Rewrite: “What is your approximate annual household income?”
- Under \$40 000 \$40 000–\$80 000 \$80 001–\$120 000 Over \$120 000
 Prefer not to say
5. (a) Observational study — the researcher observes behaviour without interfering.
- (b) A possible confounding variable is the time of day. Students who study late at night may drink more coffee and also tend to study longer regardless of the coffee.
- (c) In an observational study the researcher does not control any variables. The relationship between coffee and study time may be explained by a confounding variable. Correlation does not imply causation.
- (d) A controlled experiment: randomly assign students to two groups — one drinks coffee and one drinks a placebo (e.g. decaffeinated coffee). Measure and compare study duration for both groups under identical conditions.
- 6.
- $$\text{Sampling interval} = \frac{2\,400}{80} = 30$$
- Randomly select a starting case between 1 and 30, then inspect every 30th case after that (e.g. cases 17, 47, 77, 107, ...) until 80 cases have been checked.
- (b) If defective cases cluster at regular intervals (e.g. every 30th case from a machine fault), the systematic interval may consistently land on — or consistently miss — defective cases, biasing the results.
- (c) Stratified sampling would be more appropriate. By dividing production into strata (e.g. by shift or machine), the inspector ensures defect-prone periods are proportionally represented in the sample.
7. (a) Headline A is using the **mean**. Headline B is using the **median**.
- (b) High earners inflate the mean upward, so the mean of \$1 450 is pulled above what most workers actually earn. The median of \$1 100 is resistant to these high values and better reflects the typical worker.
- (c) The mean being much higher than the median suggests the distribution is **positively skewed** — a small number of very high earners pull the mean up while most workers earn relatively less.
- (d) Headline B gives a more accurate picture of the typical worker’s wage. The median is not distorted by high-income outliers and represents the wage that 50% of workers fall below, making it a fairer measure of the centre for skewed income data.

8. (a) Option A: systematic sampling. Option B: self-selected (convenience) sampling. Option C: self-selected sampling.
- (b) Option A: households without a listed address or on the electoral roll (e.g. renters, younger residents) would be excluded. Option B: only people who happen to visit the shopping centre are included, excluding those who shop elsewhere or are housebound. Option C: only residents with internet access and a Facebook account would see the survey, excluding older or less digitally connected residents.
- (c) Option A would produce the most representative sample. It uses systematic sampling from the electoral roll, giving all registered households a chance of selection, and is not subject to voluntary response bias.
- (d) The council should ensure responses are anonymous so residents feel safe expressing genuine opinions without fear of identification or consequence.
9. (a) The flaw is confusing **correlation with causation**. The fact that two variables are related does not mean one causes the other.
- (b) Hot weather — in summer, more people buy ice cream *and* more people swim, which increases the risk of drowning. Hot weather is the common cause of both.
- (c) A **confounding variable** (also called a lurking variable).
- (d) Example: suburbs with more hospitals tend to have higher death rates. Hospitals do not cause death — they are located where populations are larger and older.
10. (a) Participants may be unwilling to honestly disclose dietary habits (e.g. excessive fast food consumption) if they feel judged. The researcher should make the survey **anonymous** and clearly state that responses are confidential and used only for research purposes.
- (b) People who do not go to a gym would be underrepresented — they may have less healthy diets than regular gym-goers, meaning results would be biased toward healthier eating habits. People who work weekday mornings would also be excluded, potentially leaving out a large portion of full-time workers.
- (c) **Fruit consumption:** “How many serves of fruit do you eat on a typical day?”
 - None 1 serve 2 serves 3 or more serves**Fast food consumption:** “How many times per week do you eat fast food (e.g. takeaway meals)?”
 - Never 1–2 times 3–4 times 5 or more times
- (d) The sample was collected only at a gym on weekday mornings, so it is not representative of all Australian adults aged 18–65. Additionally,

the conclusion that the overall diet is healthy cannot be drawn from fruit consumption alone — other aspects of diet were not considered.